

# LOVELY PROFESSIONAL UNIVERSITY

## Sample Question Set - Unit 3

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO3

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Supervised Machine Learning

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

### Answer Grid (Key)

| Q. No. | Answer | Q. No. | Answer | Q. No. | Answer | Q. No. | Answer |
|--------|--------|--------|--------|--------|--------|--------|--------|
| Q1     | A      | Q11    | D      | Q21    | A      | Q31    | C      |
| Q2     | B      | Q12    | B      | Q22    | C      | Q32    | A      |
| Q3     | C      | Q13    | D      | Q23    | B      | Q33    | A      |
| Q4     | D      | Q14    | C      | Q24    | D      | Q34    | D      |
| Q5     | B      | Q15    | B      | Q25    | C      | Q35    | B      |
| Q6     | D      | Q16    | A      | Q26    | D      | Q36    | C      |
| Q7     | A      | Q17    | B      | Q27    | A      | Q37    | C      |
| Q8     | C      | Q18    | A      | Q28    | B      | Q38    | B      |
| Q9     | C      | Q19    | D      | Q29    | D      | Q39    | D      |
| Q10    | A      | Q20    | C      | Q30    | B      | Q40    | A      |

### Multiple Choice Questions (40 x 1 = 40 Marks)

**Q1. Name the impurity measure a decision tree may use in place of entropy.**

- A) Gini impurity <-- correct
- B) Mean squared error
- C) Cosine similarity
- D) Euclidean distance

**Answer: A.** Gini impurity and entropy both measure how mixed the classes in a node are, and in practice the two usually choose very similar splits.

**Q2. Identify the term for a model that simply stores the training data and does its work at prediction time.**

- A) An eager learner
- B) A lazy learner <-- correct
- C) An ensemble learner
- D) A generative learner

**Answer: B.** K-Nearest Neighbour fits nothing during training; every prediction requires the distances to the stored records to be computed.

**Q3. State the shape of the decision boundary that logistic regression produces in the feature space.**

- A) A closed circle around each class
- B) A curve of arbitrary shape
- C) A straight line or flat plane <-- correct
- D) A set of axis-parallel rectangles

**Answer: C.** The linear score is a weighted sum of the features, so the boundary where it crosses the threshold is linear.

**Q4. Name the sampling scheme used to build each individual tree of a Random Forest.**

- A) Stratified sampling without replacement
- B) Systematic sampling of every fifth row
- C) Cluster sampling of the training set
- D) Bootstrap sampling with replacement <-- correct

**Answer: D.** Each tree sees a different bootstrap sample, and that difference between the trees is part of what makes their average more reliable.

**Q5. Identify the value a regression tree predicts at one of its leaves.**

- A) The largest value in that leaf
- B) The mean of the training records in that leaf <-- correct
- C) The number of records in that leaf
- D) The depth of that leaf in the tree

**Answer: B.** A classification tree predicts the majority class of its leaf; a regression tree predicts the average of the target values there.

**Q6. State what the intercept of a fitted linear regression represents.**

- A) The change in the target per unit of a feature
- B) The proportion of variance the model explains

- C) The average error the model makes
- D) The predicted value when every feature is zero <-- correct

**Answer: D.** The intercept is where the fitted line meets the vertical axis. It is often not meaningful in itself when zero is outside the range of the data.

**Q7. Explain why a decision tree is described as an interpretable model.**

- A) The path from root to leaf reads as a chain of conditions <-- correct
- B) It always achieves a higher accuracy than others
- C) It requires no training data to be supplied
- D) It stores the whole training set for inspection

**Answer: A.** Each prediction can be explained by naming the tests that were applied, which is why trees are favoured where a decision has to be justified.

**Q8. Distinguish underfitting from overfitting by the pattern of the two error readings.**

- A) Overfitting has both errors high, underfitting has training low
- B) Both of the two show a high training error
- C) Underfitting has both errors high, overfitting has training low and test high <-- correct
- D) Both of the two show a low test error

**Answer: C.** The gap between the two readings is the signal: no gap and poor performance means too simple, a large gap means the model has memorised the sample.

**Q9. Describe what information gain measures at a candidate split.**

- A) The number of records the split separates
- B) The depth the tree will reach after the split
- C) The reduction in impurity the split achieves <-- correct
- D) The accuracy the finished tree will attain

**Answer: C.** The tree tries each feature and threshold and keeps the one that leaves the child nodes purest, which is the greedy step it repeats at every node.

**Q10. Interpret a fitted linear regression coefficient of - 2.5 on a feature.**

- A) The target falls by 2.5 for each unit that feature rises <-- correct
- B) The target rises by 2.5 for each unit that feature rises
- C) The feature explains 2.5 per cent of the variance
- D) The model makes an average error of 2.5 units

**Answer: A.** The sign gives the direction and the magnitude the size of the effect, both holding the other features fixed.

**Q11. Predict the effect on variance of averaging the predictions of many different trees.**

- A) The variance rises while the bias falls sharply
- B) Both the variance and the bias fall to zero
- C) Neither the variance nor the bias is affected
- D) The variance falls while the bias stays roughly the same <-- correct

**Answer: D.** Errors that are only weakly correlated cancel when averaged, which is exactly why a forest generalises better than any of its individual trees.

**Q12. Recognize the model described: it predicts the majority class among the K training records closest to the query.**

- A) Logistic regression
- B) K-Nearest Neighbour <-- correct
- C) Naive Bayes
- D) Linear regression

**Answer: B.** No model is fitted during training. The whole training set is stored and the distances are computed when a prediction is required.

**Q13. Explain why Naive Bayes performs well on text classification despite its assumption.**

- A) Words in a document really are independent
- B) Text data contains no correlated features
- C) It requires an enormous amount of training text
- D) The highest-scoring class is often right even when the probabilities are not <-- correct

**Answer: D.** The probabilities it produces are poorly calibrated, but only their ranking decides the predicted class, and it needs very little training data.

**Q14. Summarize what the bias component of a model's error describes.**

- A) Error from the model changing with the sample
- B) Error from the noise present in the measurements
- C) Error from assumptions too simple for the true pattern <-- correct
- D) Error from the features being on different scales

**Answer: C.** High bias means the model cannot represent the relationship at all, so it fits the training data and the test data equally badly.

**Q15. Distinguish a regression tree from a classification tree.**

- A) One predicts a class at its leaves, the other a number
  - B) One predicts a number at its leaves, the other a class <-- correct
  - C) Both of the two predict a class at their leaves
  - D) Both of the two predict a number at their leaves
- Answer: B.** The structure is identical; what differs is the value stored at the leaf and the criterion used to choose each split.

**Q16. Explain why a polynomial regression of very high degree fits the noise of the training data.**

A) It has enough flexibility to pass close to every point  
<-- correct

B) It reduces the number of features considered

C) It forces the relationship to be a straight line

D) It removes the extreme values from the data

**Answer: A.** Each added degree adds a bend the curve can make. With enough bends the curve chases individual observations rather than the underlying trend.

**Q17. Predict what happens to a K-Nearest Neighbour classifier when many irrelevant features are added.**

A) The distances become sharper and accuracy rises

B) The distances lose contrast and the neighbours become arbitrary <-- correct

C) The classifier automatically ignores those features

D) The classifier can no longer be trained at all

**Answer: B.** Every added feature contributes to the distance whether or not it is informative, so the near and far records come to look equally far apart.

**Q18. Describe what Laplace smoothing prevents in a Naive Bayes classifier.**

A) A single unseen word driving the whole score to zero <-- correct

B) The classifier from being trained on text data

C) The features from being treated as independent

D) The prior probabilities from being computed

**Answer: A.** Because the scores are products, one zero factor annihilates the whole product. Adding a small count to every term keeps the estimate usable.

**Q19. Explain why a Random Forest does not require its features to be placed on a common scale.**

A) It converts every feature to the same units

B) It computes distances between the records

C) It uses only the largest-ranging feature

D) Each split compares values within a single feature  
<-- correct

**Answer: D.** Thresholds are chosen inside one column at a time, so the units of the other columns never enter the comparison.

**Q20. Distinguish the parameters of a model from its hyperparameters.**

A) Hyperparameters are learned, parameters are set beforehand

B) Both of the two are learned from the training data

C) Parameters are learned from the data, hyperparameters are set beforehand <-- correct

D) Both of the two are fixed before training begins

**Answer: C.** The coefficients of a regression are parameters; the depth limit of a tree or the K of a nearest-neighbour classifier are hyperparameters.

**Q21. Apply the fitted model  $y = 3 + 1.5x$  to an input of  $x = 8$  and determine the predicted value.**

A) 15 <-- correct

B) 12

C) 13.5

D) 16.5

**Answer: A.**  $y = 3 + 1.5 \times 8 = 3 + 12 = 15$ .

**Q22. Solve the following: a regression line has a slope of 2 and passes through the point with coordinates 5 and 16. Determine its intercept.**

A) 10

B) 26

C) 6 <-- correct

D) 3.2

**Answer: C.** Substituting gives  $16 = c + 2 \times 5$ , so  $c = 16 - 10 = 6$ .

**Q23. Use a logistic regression model whose linear score for an observation is 0.8, and determine the class predicted at a threshold of 0.5.**

A) The negative class

B) The positive class <-- correct

C) Neither class

D) Both classes equally

**Answer: B.** The sigmoid of any positive score exceeds 0.5, so a positive linear score is always predicted as the positive class at this threshold.

**Q24. Apply a 5-Nearest Neighbour classifier to a query whose five closest training points carry the labels A, A, B, A and B, and determine the prediction.**

A) B

B) Either A or B at random

C) No prediction is possible

D) A <-- correct

**Answer: D.** The majority among the five neighbours is A, appearing three times against two.

**Q25. Examine a regression fit whose residuals are scattered randomly about zero with no pattern, and determine what it indicates.**

A) The model has clearly overfitted the data

B) A higher-degree polynomial is required

C) The linear form is appropriate for this data <-- correct

D) The features must be rescaled before fitting

**Answer: C.** A structureless residual plot is what a well-specified model produces. Any visible pattern means systematic error the model has failed to capture.

**Q26. Compare a decision tree limited to a depth of one with an unlimited tree on bias and on variance.**

A) The shallow tree has high variance, the unlimited tree high bias

- B) Both of the two have high bias and low variance
- C) Both of the two have low bias and low variance
- D) The shallow tree has high bias, the unlimited tree high variance <-- correct

**Answer: D.** A single split is too crude to capture the pattern; an unlimited tree isolates individual records and changes completely with a new sample.

**Q27. Solve the following: at a candidate split a node holding 60 samples divides into children of 45 and 15. Determine the fraction that goes to the larger child.**

- A) 0.75 <-- correct
- B) 0.25
- C) 0.45
- D) 0.60

**Answer: A.**  $45 / 60 = 0.75$ , the weight that child receives when the impurity of the split is computed.

**Q28. Apply Naive Bayes to a two-class problem in which the unnormalised score for class A is 0.006 and for class B is 0.024, and determine the prediction.**

- A) Class A
- B) Class B <-- correct
- C) Neither class
- D) Both are equally likely

**Answer: B.** Only the relative sizes matter, so the class with the larger score is predicted. The normalising denominator is the same for both and can be ignored.

**Q29. Use a learning curve on which the training error is very low while the validation error remains high, and determine the diagnosis.**

- A) The model is underfitting the training data
- B) The model has an ideal level of complexity
- C) The training set is far too large for the model
- D) The model is overfitting the training data <-- correct

**Answer: D.** The model reproduces the training rows including their noise. Reducing complexity, adding data or regularising are the usual responses.

**Q30. Compute the R squared of a regression whose residual sum of squares is 20 and whose total sum of squares is 100.**

- A) 0.2
- B) 0.8 <-- correct
- C) 5.0
- D) 0.02

**Answer: B.**  $R^2$  is  $1 - 20 / 100 = 1 - 0.2 = 0.8$ , so the model accounts for eighty per cent of the variation in the target.

**Q31. Apply a Random Forest of 100 trees in which 60 vote Faulty and 40 vote Healthy, and determine the prediction.**

- A) Healthy

- B) Neither of the two
- C) Faulty <-- correct
- D) The vote is a tie

**Answer: C.** For a classification target the forest predicts by majority vote; for a regression target it averages the values the trees produce.

**Q32. Examine a K-Nearest Neighbour classifier with K set to 1 on noisy data and determine its behaviour.**

- A) A jagged boundary that follows the noise <-- correct
- B) A smooth boundary that ignores the noise
- C) A straight boundary through the feature space
- D) A prediction of the majority class everywhere

**Answer: A.** With a single neighbour each training point governs its own region, so the boundary has very high variance. Raising K smooths it at the cost of bias.

**Q33. Apply polynomial regression of degree one and determine what model it is equivalent to.**

- A) Ordinary linear regression <-- correct
- B) A decision tree regressor
- C) A K-Nearest Neighbour regressor
- D) A logistic regression classifier

**Answer: A.** A first-degree polynomial is a straight line, so the two are the same model. Degrees above one add the bends that let the curve follow non-linear shapes.

**Q34. Use the sigmoid function to determine the predicted probability when the linear score is exactly zero.**

- A) 0
- B) 1
- C) It is undefined
- D) 0.5 <-- correct

**Answer: D.** The sigmoid maps zero to one half, which is why the boundary of a logistic classifier at a threshold of 0.5 is the set of points with a zero score.

**Q35. Compare the Gini impurity of a node holding one class alone with that of a node split evenly between two classes.**

- A) 0.5 for the pure node and zero for the even one
- B) Zero for the pure node and 0.5 for the even one <-- correct
- C) Zero for both of the two nodes
- D) One for both of the two nodes

**Answer: B.** A pure node has no impurity to reduce; an evenly split node has the greatest impurity a two-class node can hold.

**Q36. Apply bootstrap sampling to a training set of 100 records and determine the size of each sample drawn.**

- A) 50 records, drawn without replacement
- B) 100 records, drawn without replacement

- C) 100 records, drawn with replacement <-- correct
- D) 10 records, drawn with replacement

**Answer: C.** Each bootstrap sample is the same size as the original but contains repeats, so roughly a third of the records are left out of any given tree.

**Q37. Examine a model recording 95 per cent accuracy on the training set and 94 per cent on the test set, and determine what it indicates.**

- A) The model has badly overfitted the training set
- B) The model has badly underfitted the training set
- C) The model generalises well to unseen data <-- correct
- D) The test set must have leaked into training

**Answer: C.** A small gap between the two readings, with both of them high, is the pattern a well-fitted model produces.

**Q38. Use the bias-variance idea to determine the effect of steadily increasing the complexity of a model.**

- A) Bias rises while variance falls
- B) Bias falls while variance rises <-- correct
- C) Both bias and variance fall together
- D) Both bias and variance rise together

**Answer: B.** The total error first falls as bias is reduced, then rises as variance takes over. The best complexity sits at the bottom of that curve.

**Q39. Apply a decision threshold of 0.5 to the predicted probabilities 0.72, 0.31 and 0.55, and determine how many records are called positive.**

- A) 1
- B) 3
- C) 0
- D) 2 <-- correct

**Answer: D.** Only 0.72 and 0.55 exceed the threshold, so two records are labelled positive.

**Q40. Relate the number of trees in a Random Forest to the stability of the predictions it makes.**

- A) More trees give steadier predictions with diminishing returns <-- correct
- B) More trees make the predictions less steady
- C) The number of trees has no effect on stability
- D) More trees always raise the accuracy without limit

**Answer: A.** Averaging more trees reduces the variance of the ensemble, but the improvement flattens off while the cost of training and prediction keeps rising.

----- End of Sample Question Set -----